

tions requiring quick power discharge, they are used in energy harvesting, automated guided vehicles (AGV), electric vehicle power, IoT devices, and emergency lighting. They also have internal cell balancing, high energy density, low self-discharge rates, and robust durability, enduring over 500,000 charge/discharge cycles with a lifespan up to 10 years.

Knowles Precision Devices
<https://www.knowlesc capacitors.com>

700-1300V power capacitors

TDK Corporation has introduced the B25695E series of EPCOS power capacitors, designed for DC link applications



in high-temperature environments up to +105°C. These capacitors operate with rated DC voltages from 700V to 1300V and require voltage derating above +85°C. They have a 35% longer lifespan than standard versions, with a projected service life of 120,000 hours at +75°C. Suited for renewable energy, rolling stock, and industrial drives, these RoHS-compliant capacitors provide efficient energy storage and enhanced system performance and reliability.

TDK Corporation
<https://www.tdk.com/en>

60MP smartphone image sensor

OMNIVISION's new OV50K40 is a 50-megapixel smartphone image sensor featuring TheiaCel technology,



which enhances high dynamic range (HDR) using lateral overflow integration capacitors for

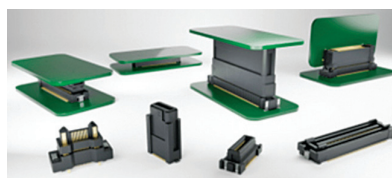
superior image quality in varying light conditions. This sensor includes 1.2-micron pixels on a 1/3.3cm (1/1.3-inch) format with high gain and multiple sampling for optimal low-light performance. Additionally, it features quad-phase detection for precise autofocus, and supports high-

resolution 50MP images, 8K video, and 2x crop-zoom. Built with PureCelPlus-S technology, the OV50K40 advances mobile photography, especially in challenging lighting.

OMNIVISION
<https://www.ovt.com>

Data and power transfer connectors

Harwin has launched the Flecto family of board-to-board connectors, designed for high-performance applications involving data and power

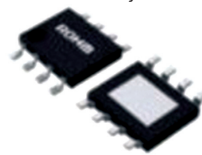


transfer. These connectors, accommodating up to 160 contacts, are available in three miniature pitch sizes and are capable of handling small multidirectional movements. The connectors are available for both vertical and horizontal configurations, facilitating parallel or perpendicular PCB connections, enhancing their application in sectors like factory automation, electric vehicles, and IoT devices. The connectors address the growing need for precision and reliability in environments prone to mechanical misalignment and stress.

Harwin
<https://www.harwin.com>

Automotive LDO regulators

ROHM has launched a new series of primary LDO regulators, the BD9xxM5-C, to address the automotive industry's needs for reliable



power supplies amid the shift towards electrification. These regulators are

essential for stepping down battery voltage to power MCUs and other components in electronic control units (ECUs), especially against the back-

drop of frequent voltage fluctuations in vehicle batteries. They are suitable for a variety of applications, including powertrain systems, body control modules, and infotainment systems, and facilitate easier power circuit design with minimal output capacitance.

ROHM
<https://rohm.com>

Automotive step-up switching regulator controllers

ABLIC has released the S-19990/9 Series of automotive step-up switching regulator controllers, designed to enhance the reliability of electronic control units (ECUs) in vehicles. As



vehicles advance in electrification, ECUs—critical for systems like electric door latches and emergency calls—

often rely on a 12V auxiliary battery. The new controllers are essential for maintaining ECU function in accidents by converting low-voltage capacitor or battery power to 12V, ensuring operational continuity. These 36V controllers start at 3V and come in oscillation frequencies of 400kHz or 2.2MHz.

ABLIC
<https://www.ablic.com/en/semicon>

EMBEDDED

Pre-configured COM and cloud modules

Congatec has introduced aReady.COM, a new product line under its aReady strategy aimed at simplifying the adop-



tion of embedded and edge computing technologies. This lineup features pre-configured

computer-on-modules (COM) and cloud solutions to ease the integration process for OEMs and developers. These modules are equipped with ready-to-use hypervisors, operating systems, and IIoT software, allowing